

Rupendra Pratap Singh Hada

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Researcher specialising in AI-driven IoT and edge computing systems for real-world intelligent environments, with a focus on anomaly-aware sensing, energy-efficient wireless networks, and scalable data-driven architectures. My work bridges sensor-level intelligence and system-level decision-making, contributing to smart environments, digital twin systems, and sustainable infrastructure. I have demonstrated this through real-world deployments and high-impact publications in ACM Transactions journals. My research aims to develop sustainable, resilient, and citizen-centric intelligent systems for real-world societal impact.

PUBLICATIONS

My research contributions focus on scalable IoT systems, anomaly-resilient sensor networks, and edge-enabled intelligent environments, and have been published in leading ACM Transactions and SCI-indexed journals.

• Journals

1. **R. P. S. Hada** and A. Srivastava, "A hybrid approach for localisation of sensor nodes in remote locations," *ACM Transactions on Sensor Networks (SCI, Q1)*, vol. 21, no. 2, pp. 1–33, 2025.
DOI: <https://doi.org/10.1145/3715914>
2. **R. P. S. Hada** and A. Srivastava, "Dynamic cluster head selection in WSN," *ACM Transactions on Embedded Computing Systems (SCI, Q2)*, vol. 23, no. 4, pp. 64:1–64:27, 2024.
DOI: <https://doi.org/10.1145/3665867>
3. **R. P. S. Hada**, U. Aggarwal, and A. Srivastava, "A study and analysis of a new hybrid approach for localisation in wireless sensor networks," *Journal of Web Engineering, (SCI, Q3)*, vol. 22, no. 2, pp. 279–302, 2023.
DOI: <https://doi.org/10.13052/jwe1540-9589.2224>
4. **R. P. S. Hada** and A. Srivastava, "Priority-based scheduler for asymmetric multi-core edge computing," *Journal of Web Engineering, (SCI, Q3)*, vol. 22, no. 6, pp. 871–888, 2023.
DOI: <https://doi.org/10.13052/jwe1540-9589.2262>
5. **R. P. S. Hada**, "CMA-ES-SVR based localisation in indoor environments," *ACM SenSys*, Submitted.

• Conferences

1. **R. P. S. Hada**, and A. Srivastava, "A novel priority-based scheduler for asymmetric multi-core edge computing," in *International Conference on Web Engineering, Springer, (Core B*)*, 2023, pp. 7–18.
DOI: https://doi.org/10.1007/978-3-031-50385-6_1

• In Submission

1. **R. P. S. Hada**, "Energy-Aware Federated Learning for Communication-Efficient Localisation in Wireless IoT Networks," *IEEE Transactions on Communications, (SCI, Q1)*.
2. **R. P. S. Hada**, "Heuristic based dynamic Cluster Head selection in WSN," *ACM Transactions on Internet Technology, (SCI, Q1)*.

RESEARCH GRANTS & FUNDED PROJECTS

• Defries–Bajpai Foundation Research Grant (USA)

Aug 2021 - July 2025

Energy-Efficient Wireless Sensor Networks & Embedded Systems

India

- **Role:** Doctoral Researcher
- Conducted research on energy-aware clustering and large-scale localisation in WSNs.
- Resulted in publications in **ACM Transactions on Sensor Networks** (2025) and **ACM Transactions on Embedded Computing Systems** (2024).

• INTPART DTRF Project – Research Council of Norway (Grant No. 309448)

May 2023 - July 2024

Indian–Norwegian Collaboration on Distributed & IoT Systems

Norway

- **Role:** Research Contributor / Visiting Researcher
- Contributed to a competitive international research project (INTPART DTRF, Research Council of Norway, Grant No. 309448) on scalable localisation and edge-enabled IoT architectures, collaborating with international research teams.
- Published research under this grant (**ACM Transactions on Internet Technology – Under Review**).

RESEARCH

- **AI-Driven and Anomaly-Resilient Network Optimization in IoT Systems**
 - Designed intelligent clustering mechanisms for large-scale IoT networks under dynamic and uncertain conditions.
 - We proposed a hybrid K-Means + SOM framework with multi-factor optimization (energy, connectivity, spatial distribution), improving network lifetime and robustness. This work addresses challenges of data variability and anomaly-prone environments in real-world sensor deployments.
- **Scalable Localisation for Real-World Environmental Monitoring Systems**
 - Developed a hybrid RSSI + AoA-based localisation framework for large-scale outdoor IoT deployments, reducing dependency on anchor nodes to 1%.
 - Implemented and validated using a low-cost prototype, enabling reliable sensing in sparse and uncertain environments. This work supports real-world applications such as environmental monitoring and anomaly detection in smart ecosystems.
- **Edge Intelligence for Resource-Constrained IoT Systems**
 - Designed a priority-aware scheduling framework for asymmetric multi-core edge devices, enabling efficient execution of latency-critical tasks.
 - Achieved up to 16% performance improvement over standard Linux scheduling. This work contributes to real-time edge intelligence required for scalable digital twin systems and distributed AI.
- **Hybrid Recommendation Model for Cold-Start Problem**
 - Cold-start scenarios limit recommendation accuracy due to insufficient user history data.
 - We proposed a four-phase model integrating frequent pattern mining, KNN-based behavioral analysis, and cost/brand filtering, achieving high prediction accuracy of 90% with reduced computational overhead in a Java-based implementation.

These research contributions collectively address challenges in real-world IoT systems, including data uncertainty, scalability, and anomaly-prone environments, forming the foundation for my current work on anomaly-aware and human-centric digital twin systems.

FUTURE RESEARCH DIRECTIONS

- **Anomaly-Aware and Human-Centric Digital Twins**
- **Behaviour-Aware AI for Smart Cities and Rural Systems**
- **Sustainable and Resilient Smart Environments**
- **Trustworthy and Explainable AI for IoT Systems**

EDUCATION

- **Indian Institute of Technology Indore** Aug 2021 - July 2025
Ph.D. in Computer Science and Engineering Indore, India
 - Advisor: Prof. Dr. Abhishek Srivastava, CGPA: 9.0
 - Thesis title: Energy Preservation in Wireless Sensor Networks
 - The doctoral thesis resulted in 4 SCI publications.
- **Shri G. S. Institute of Technology and Science** July 2016 - April 2019
M.Tech in Computer Science and Engineering Indore, India
 - Advisor: Prof. D.A. Mehta, CGPA: 7.46
 - Thesis title: Parallel Priority-Based Scheduling on Asymmetric Multicore Processors
- **Rajiv Gandhi Proudyogiki Vishwavidyalaya** July 2011 - May 2015
B.Tech in Computer Science and Engineering Bhopal, India

WORK EXPERIENCE

- **Indian Institute of Technology** July 2025 - till
Research Associate Indore, India
 - Working on AI-driven IoT systems for resource-constrained environments, focusing on anomaly detection, edge intelligence, and scalable data processing for real-world deployments.
- **Western Norway Research Institute** May 2024 - July 2024
Visiting Researcher Sogndal, Norway
 - Worked on INTPART DTRF project (by Research Council of Norway and the Norwegian Directorate for Higher Education and Skills (HK-dir)).
 - Contributed to international research on scalable IoT architectures and distributed intelligence, focusing on real-world deployments and data-driven system design in collaboration with global research teams.
- **Indian Institute of Technology Indore** Aug 2021 - July 2025
Teaching & Research Assistant, Department of Computer Science and Engineering Indore, India
 - Designed and delivered tutorials, laboratory modules, and assessments for undergraduate and graduate courses, including **Computer Programming, CS-103, Advanced Algorithms, CS-411/611, and Data Structures & Algorithms, CS-303** (serving 450+ students).
 - Contributed to curriculum development and project evaluation for advanced-level courses.

ACADEMIC SERVICE

- Reviewer – *Discover Artificial Intelligence* (Springer Nature, Q1).
- Reviewer – *Cluster Computing* (Springer Nature, Q1).
- Reviewer – *Journal of Supercomputing* (Springer, Q2).
- Reviewer – *Wireless Networks* (Springer Nature, Q2).

SKILLS

- **Programming:** Python, C/C++, R, MATLAB
- **Data Analysis:** Time-series analysis, anomaly detection, sensor data processing
- **Machine Learning:** PyTorch, TensorFlow
- **Systems & Tools:** Linux, Docker, Git
- **IoT & Embedded Systems:** Arduino, Sensor deployment

POSITION OF RESPONSIBILITIES & EXTRACURRICULAR ACTIVITIES

- **Mentorship:** Supervised 8 undergraduate and 3 post-graduate students in their research projects.
- **Overall Co-ordinator:** Symposium 2.0, IIT Indore – Organised a technical symposium on emerging CSE trends, engaging 150+ students and faculty.
- **Head Co-ordinator:** CSE Events & Outreach, IIT Indore – Managed 20+ academic and career-oriented events including seminars, symposia, and outreach activities.
- **Education & Skill Development Lead:** Led NGO initiatives in rural Madhya Pradesh, impacting 500+ students through health, education, and vocational programs.